

**LESSON 2-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**

# Median and Outliers

Maryland Standard 6.SP.B.5d • Learning target – I can explain why the median is a better measure of center than the mean when a data set has an outlier.

**REMEMBER**

- An **outlier** is a value much greater or much less than the rest of the data.
- An outlier pulls the **mean** toward it. The **median** barely moves, so it is **resistant** to outliers.
- With an outlier, the median usually describes a **typical value** better. Still report the outlier; it is real data.

**WORKED EXAMPLE**

Six bike rides lasted 14, 18, 15, 16, 64, and 17 minutes. The mean is 24 minutes. Which measure better describes a typical ride?

**Step 1** Order: 14, 15, 16, 17, 18, 64. The outlier is 64.

**Step 2** Median:  $(16 + 17) \div 2 = 16.5$ . The mean, 24, is greater than 5 of the 6 rides.

**Answer** The median, 16.5 minutes

**WORD BANK**

**outlier** – a number far away from the others

**median** – the middle number

**mean** – add all, then divide by how many

**typical** – usual, like most of the data

**PRACTICE**

*show your work • label units*

1. Find the median of each data set. Circle the outlier in each set.

**Step 1** Order each list.

**Step 2** Circle the number far from the rest.

**Step 3** Find the middle number.

a. 9, 11, 10, 12, 13, 8, 48 \_\_\_\_\_

b. 3, 70, 72, 75, 78, 71, 74 \_\_\_\_\_

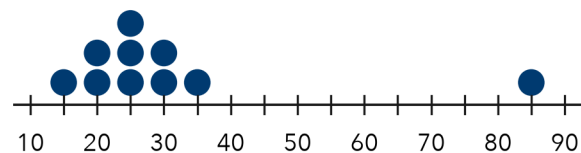
c. 25, 28, 30, 31, 33, 95 \_\_\_\_\_

d. 40, 41, 43, 44, 46, 47, 150 \_\_\_\_\_

e. 6, 7, 7, 8, 9, 10, 34 \_\_\_\_\_

f. 56, 52, 55, 54, 9, 53, 57, 58 \_\_\_\_\_

- 2. Homework Minutes.** The dot plot shows minutes 10 students spent on homework one night. The mean is 31 minutes.



**Step 1** Find the dot far from the others.

**Step 2** Order the 10 values; the median is between the 5th and 6th.

**Step 3** Count students below 31.

a. Outlier: \_\_\_\_\_

b. Median: \_\_\_\_\_ min

c. Students below the mean:

\_\_\_\_\_

d. Better center: \_\_\_\_\_

- 3. Library Visitors.** Visitors on 7 days: 64, 58, 61, 66, 59, 63, and 4 (the library closed early for snow).

a. Median of all 7 days: \_\_\_\_\_

b. Median without the 4: \_\_\_\_\_

- 4. Car Wash.** A club earned \$10, \$12, \$13, \$14, \$15, \$16, and \$60 on 7 Saturdays.

**Step 1** Mean: add the 7 amounts, then divide by 7.

**Step 2** Median: the 4th amount in order.

a. Mean: \$ \_\_\_\_\_

b. Median: \$ \_\_\_\_\_

c. Better center: \_\_\_\_\_

- 5. Find the Error.** Andre's data are 31, 33, 34, 36, 37, 39, and 210. He says, "The outlier 210 makes the median much bigger, so the median is not a good measure." Is he right? Find the median to support your answer.

**Frame** Andre is \_\_\_\_\_. The median is \_\_\_\_\_. Without 210 it would be \_\_\_\_\_.

\_\_\_\_\_

\_\_\_\_\_

**CHECK YOURSELF**

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #\_\_\_\_\_

**AT HOME** Ask someone at home about a normal trip and their slowest trip. Which trip is an outlier?

**LESSON 2-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2****Comparing Distributions**

Maryland Standard 6.SP.B.5d • Learning target – I can compare two distributions using the median and the interquartile range together.

**REMEMBER**

- Compare **centers** with the medians. Compare **spreads** with the IQRs ( $IQR = Q3 - Q1$ ).
- A smaller IQR means more **consistent** data. A full comparison names **both** measures for **both** groups, with numbers.

**WORKED EXAMPLE**

Team Red: median 14 points, IQR 6. Team Blue: median 18 points, IQR 15. Compare the teams.

**Step 1** Centers:  $18 > 14$ , so Blue usually scores more.

**Step 2** Spreads:  $6 < 15$ , so Red is more consistent.

**Answer** **Blue scores more; Red is more consistent.**

**WORD BANK**

**median** – the typical value

**IQR** –  $Q3 - Q1$ , the spread of the middle half

**consistent** – close together

**predictable** – you can guess it well

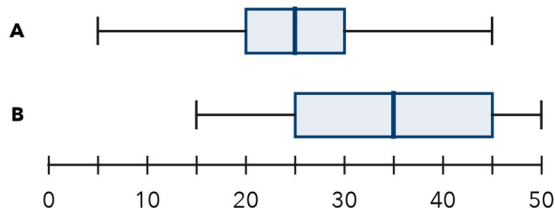
**PRACTICE**

*show your work • label units*

- 1. Reading Minutes.** The box plots show minutes read per day by Class A and Class B.

**Step 1** Find the line in each box: the medians.

**Step 2**  $IQR = \text{right edge} - \text{left edge}$  of each box.



- Medians: A \_\_\_\_ B \_\_\_\_
- IQRs: A \_\_\_\_ B \_\_\_\_
- Usually reads more: \_\_\_\_\_
- More consistent: \_\_\_\_\_

- 2. Bus Routes.** Minutes late on 7 days. **Route 3:** 2, 3, 5, 6, 8, 9, 11. **Route 8:** 0, 1, 4, 6, 10, 14, 15.

**Step 1** Order is given. The median is the 4th value.

**Step 2** Q1 is the 2nd value, Q3 is the 6th value.

- Route 3: median \_\_\_\_ IQR \_\_\_\_ Route 8: median \_\_\_\_ IQR \_\_\_\_
- Which route is more predictable? Why? \_\_\_\_\_

- 3. Plant Heights (cm).** **Sun:** 10, 12, 13, 15, 16, 18, 19, 21. **Shade:** 6, 7, 8, 8, 9, 10, 10, 12.

- Sun: median \_\_\_\_ cm IQR \_\_\_\_ cm
- Shade: median \_\_\_\_ cm IQR \_\_\_\_ cm

4. **Delivery.** Shop K: median 28 min, IQR 8 min. Shop L: median 36 min, IQR 8 min. What is the same?

☐ the same center                      ☐ the same spread                      ☐ both center and spread

5. **Find the Error.** Set E: median 70, IQR 25. Set F: median 64, IQR 6. Jada says, "Set E is higher and more consistent." Explain her error.

**Frame** Set E has the higher \_\_\_\_\_. The more consistent set is \_\_\_\_\_ because its \_\_\_\_\_ is \_\_\_\_\_.

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## TEST PRACTICE

*choose the best answer*

6. Group 1: median 12 cm, IQR 3 cm. Group 2: median 12 cm, IQR 9 cm. Which is more consistent?

**A** Group 1                      **B** Group 2                      **C** Both the same                      **D** Cannot tell

7. What is the IQR of 3, 5, 6, 8, 10, 12, 13?

**A** 7                      **B** 8                      **C** 10                      **D** 17

### CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_

**AT HOME** Tell someone at home why a bus with a small IQR is easier to plan around.